

DocuMind AI — Product Documentation

v1.0

Overview

DocuMind AI is an enterprise-grade Retrieval-Augmented Generation (RAG) system that enables organisations to query their internal documents using natural language. The system combines state-of-the-art LLMs with vector search to deliver accurate, cited answers from your document corpus.

Key capabilities: multi-document Q&A, real-time streaming responses, source citations with page numbers, conversation memory, and a REST API.

1. System Requirements

- Python 3.10 or higher
- RAM: Minimum 8 GB (16 GB recommended for large document sets)
- Disk: 10 GB free space for models and vector indexes
- OS: Ubuntu 20.04+, macOS 12+, Windows 11
- GPU: Optional — NVIDIA GPU with CUDA 11.8+ improves embedding speed by 10x
- Internet: Required for OpenAI API (optional if using local models)

2. Installation

Step 1 — Clone the repository: `git clone https://github.com/yourusername/documind-ai.git` `cd documind-ai`

Step 2 — Create virtual environment: `python -m venv venv` `source venv/bin/activate` # Linux/Mac
`venv\Scripts\activate` # Windows

Step 3 — Install dependencies: `pip install -r requirements.txt`

Step 4 — Configure environment: `cp .env.example .env` # Edit .env with your API keys

Step 5 — Generate sample data: `python generate_sample_docs.py`

Step 6 — Run the dashboard: `streamlit run dashboard/app.py`

3. Configuration Guide

All configuration is managed via `config/config.yaml`. Key settings:

- `llm.provider`: 'openai' for GPT models, 'huggingface' for free models

- llm.model_name: 'gpt-3.5-turbo' (default) or 'gpt-4' for higher quality
- llm.temperature: 0.0 for factual responses, 0.7 for creative responses
- embeddings.provider: 'openai' (ada-002) or 'huggingface' (MiniLM, free)
- vector_store.top_k: Number of document chunks to retrieve (default: 5)
- chunking.chunk_size: Text chunk size in characters (default: 1000)
- retrieval.search_type: 'mmr' for diverse results, 'similarity' for pure relevance
- FREE_MODE=true in .env enables fully local operation at no cost

4. API Reference

The REST API is available at <http://localhost:8000>. Base path: /api/v1

4.1 Chat Endpoints

POST /api/v1/chat Request body: {"question": "string", "session_id": "string", "stream": bool} Response: {"answer": "string", "sources": [], "confidence": float, "tokens_used": int, "response_time": float}

POST /api/v1/chat/reset Request body: {"session_id": "string"} Clears conversation history for the given session.

GET /api/v1/chat/history/{session_id} Returns the full conversation history with metadata.

4.2 Document Endpoints

POST /api/v1/documents/upload Multipart form-data. Accepts PDF, DOCX, TXT, CSV, PPTX. Response: {"doc_id": str, "chunks_created": int, "processing_time": float}

GET /api/v1/documents/list Returns all indexed documents with metadata.

DELETE /api/v1/documents/{doc_id} Removes a document and its vectors from the index.

GET /api/v1/documents/stats Returns total vectors, index size, and per-document statistics.

5. Troubleshooting

Common issues and their resolutions:

- Error 'No module named faiss': Run 'pip install faiss-cpu' (or faiss-gpu for GPU)
- OpenAI AuthenticationError: Verify OPENAI_API_KEY in your .env file
- Slow embedding speed: Set FREE_MODE=false to use OpenAI Ada-002 (much faster)
- Empty answers: Lower score_threshold in config.yaml (default 0.3 → try 0.1)

- Memory issues with large PDFs: Reduce `chunk_size` to 500 in `config.yaml`
- Port 8501 already in use: Kill existing Streamlit with `'pkill -f streamlit'`
- FAISS index corrupt: Delete `data/processed/embeddings/` and re-ingest documents
- Rate limit from OpenAI: Implement exponential backoff or switch to `gpt-3.5-turbo`

6. FAQ

- Q: Can I use DocuMind AI without an OpenAI API key? A: Yes, set `FREE_MODE=true`.
- Q: What is the maximum document size? A: 50 MB per file (configurable).
- Q: How many documents can I index? A: FAISS supports millions of vectors.
- Q: Is my data sent to OpenAI? A: Only if using OpenAI APIs; `FREE_MODE` keeps data local.
- Q: Can I use my own embedding model? A: Yes, set `embeddings.provider` to `'huggingface'`.
- Q: How do I improve answer quality? A: Increase `top_k`, use `gpt-4`, reduce `chunk_size`.
- Q: Does it support multiple languages? A: Yes, but best results with English documents.